

Euclid's Elements — Book I

Proposition 11

Formal Reconstruction — Technical Documentation

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Constructing a perpendicular at a point on a line

1 Introduction

Proposition I.11 states:

To draw a straight line at right angles to a given straight line from a prescribed point on it.

Let AB be the given straight line and let C be the prescribed point on AB .

Euclid's construction is concrete. Choose a point D on one side of C , lay off on the opposite ray a point E with

$$CD \cong CE,$$

construct the equilateral triangle DFE , and join C to F . SSS for $\triangle DCF$ and $\triangle ECF$ gives equality of the adjacent angles at C . Since $D - C - E$, Euclid's Definition 10 identifies those equal adjacent angles as right angles.

The current Hilbert reconstruction proves the same existence statement by a different synthetic mechanism. The public theorem in `Proposition11.lean` is only an interface theorem; the substantial geometry lives in the Book Zero theorem `hilbert_right_angle_exists_nondegenerate`.

2 The Classical Configuration

The classical configuration is

$$D - C - E, \quad CD \cong CE, \quad DF \cong FE,$$

with DFE equilateral and CF joined. The proof compares

$$\triangle DCF \quad \text{and} \quad \triangle ECF.$$

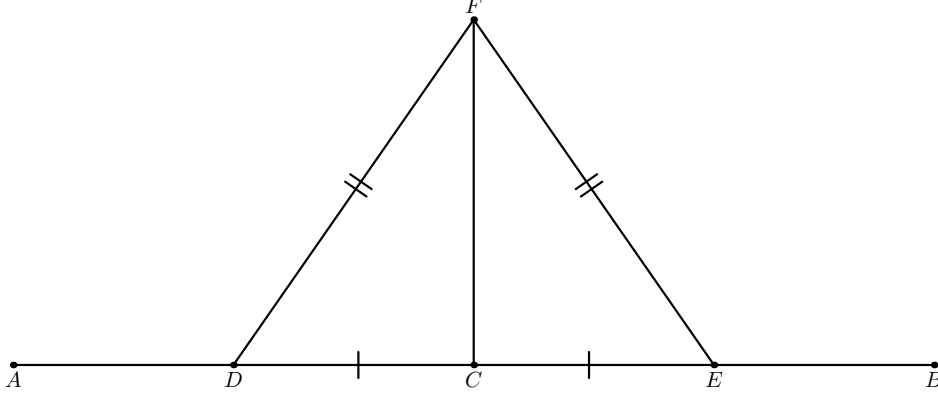


Figure 1: Classical Euclid–Wilkins construction for Proposition I.11. The points D, C, E are collinear with C between D and E and $CD \cong CE$; DFE is equilateral. The segment CF is the candidate perpendicular.

3 Statement

The synthetic content of Proposition I.11 is the existence of a right angle through a prescribed point of a straight line.

Theorem 1 (Euclid I.11). *Let A, C, B satisfy*

$$A * C * B.$$

Then there exists a point X such that

$$\text{HilbertRightAngle}(A, C, X).$$

The right-angle predicate is the synthetic adjacent-angle formulation:

$$\text{HilbertRightAngle}(A, O, B)$$

means that there exists a point C' such that

$$A * O * C'$$

and

$$\angle AOB \cong \angle BOC'.$$

No numerical value such as 90° is used.

The current proposition file has the form

```
theorem euclid_proposition_11
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  Exists (fun X : Geo.Point =>
    HilbertRightAngle Geo A C X) := by
exact
  hilbert_right_angle_exists
    Geo A C B hACB
```

Thus the theorem carrying Euclid's name is a thin wrapper.

4 Classical Proof

Choose D on one side of C on the given line. By Proposition I.3, lay off on the opposite ray a point E such that

$$CD \cong CE.$$

Hence

$$D - C - E.$$

By Proposition I.1 construct the equilateral triangle DFE and join C to F ; see Figure 1. Then

$$DF \cong EF, \quad CF \cong CF.$$

Together with $CD \cong CE$, Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

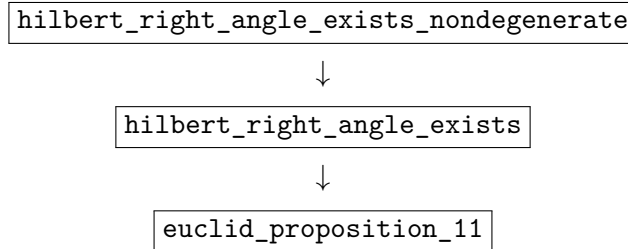
Since CD and CE are opposite rays, the two congruent angles are adjacent angles on a straight line. By Euclid's Definition 10 they are right angles. Therefore CF is perpendicular to the given line at C .

The classical dependency pattern is therefore

$$\text{I.3} \longrightarrow \text{segment layoff} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow \text{I.11}.$$

5 Current Formal Architecture

The current production architecture has three theorem levels:



The first theorem proves the stronger conclusion

$$\exists X, \quad \neg \text{Collinear}(A, C, X) \quad \wedge \quad \text{HilbertRightAngle}(A, C, X).$$

The second theorem discards the explicit noncollinearity component and returns only the right-angle witness. The proposition theorem is then a direct wrapper over that result.

This corrects an older description in which `hilbert_right_angle_exists` was presented as the theorem that carried the entire construction itself.

6 Hilbert Reconstruction

Assume

$$A * C * B.$$

Let b be the line through A and B , so $C \in b$.

6.1 Off-line point and angle construction

Choose an arbitrary point $D \notin b$. Hilbert III.4 is then applied at C to copy the angle ACD to the ray based on CB , on the side of b selected by D . It produces a point E with

$$\text{SameSide}(E, D; b)$$

and

$$\angle ACD \cong \angle BCE.$$

The current source also returns the uniqueness field of the angle construction theorem, but that field is not consumed by the active proof of I.11. The proof uses the side condition and the angle congruence.

At this point the proof performs its first genuine geometric case split:

$$\text{Collinear}(C, E, D) \quad \text{or} \quad \neg \text{Collinear}(C, E, D).$$

The four formal states needed for the audit are collected in Figure 2.

6.2 Collinear branch

Suppose C, E, D are collinear. Because E and D lie on the same side of the original base line, they cannot lie on opposite rays from C . The proof obtains

$$\text{SameRay}(C; E, D).$$

Hence the copied angle congruence transports to

$$\angle ACD \cong \angle BCD$$

and then, by reversal of the second angle,

$$\angle ACD \cong \angle DCB.$$

Since $A - C - B$, the point B supplies the extension witness in the definition of a right angle. Thus this branch ends immediately with

$$\text{HilbertRightAngle}(A, C, D).$$

In particular, the output point in this branch is simply $X = D$.

6.3 Noncollinear branch: layoff and midpoint

Now suppose

$$\neg \text{Collinear}(C, E, D).$$

On ray CD , segment construction produces F with

$$CF \cong CE.$$

Then midpoint existence produces X satisfying

$$F - X - E, \quad FX \cong XE.$$

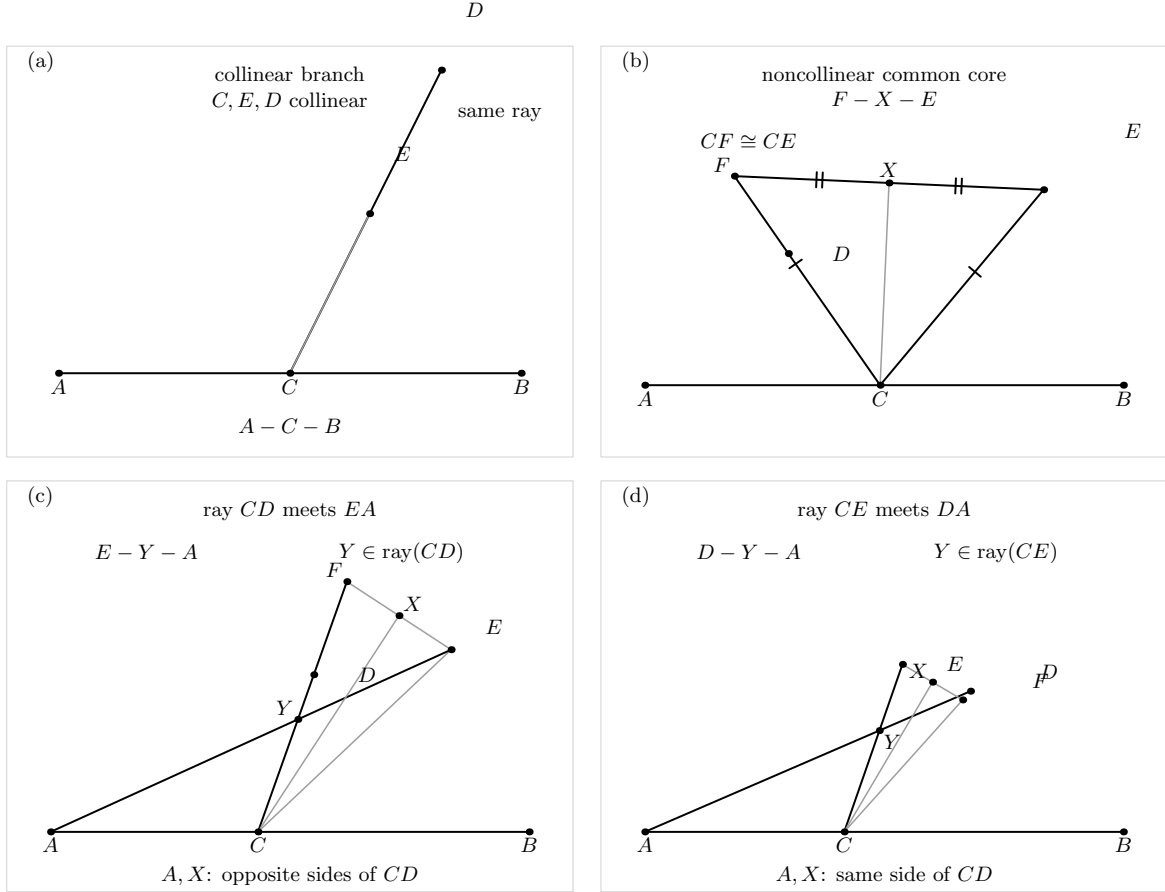


Figure 2: Formal Hilbert geometry used by the current I.11 proof. (a) If C, E, D are collinear, the same-side information forces CE and CD to be the same ray, and D itself is returned as the right-angle point. (b) In the noncollinear branch, F is laid off on ray CD with $CF \cong CE$, and X is the midpoint of FE . Panels (c) and (d) are the two subsequent ray-order cases: ray CD meets the open segment EA , or ray CE meets the open segment DA . The relative locations are qualitative representatives of the proved incidence and side data; no additional point order is asserted beyond the labelled betweenness relations.

Hilbert SSS applied to

$$\triangle CFX \quad \text{and} \quad \triangle CEX$$

gives

$$\angle FCX \cong \angle ECX.$$

Since F and D are on the same ray from C , this becomes

$$\angle DCX \cong \angle ECX.$$

Together with the angle-copy relation

$$\angle ACD \cong \angle BCE,$$

these are the two component congruences to be assembled by angle addition.

6.4 Ray-order case split

The noncollinear configuration still does not determine automatically how the component angles are arranged. The proof therefore invokes the ray-order theorem and obtains exactly two possibilities:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case there is a point Y with

$$E - Y - A$$

and Y on ray CD . The SameSide/OppositeSide calculus proves that A and X are on opposite sides of line CD , while B and X are on opposite sides of line CE .

In the second case there is a point Y with

$$D - Y - A$$

and Y on ray CE . Here the corresponding pairs are on the same sides of their reference lines.

Both branches therefore establish the invariant

$$\text{SameSide}(A, X; CD) \iff \text{SameSide}(B, X; CE).$$

This is precisely the side configuration required by the synthetic angle-addition theorem.

6.5 Angle addition and right angle

Angle addition combines

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX$$

to obtain

$$\angle ACX \cong \angle BCX.$$

After reversing the second angle,

$$\angle ACX \cong \angle XCB.$$

The original hypothesis $A - C - B$ means that the existing point B is already the extension witness required by `HilbertRightAngle`. No further geometric construction is needed at the endpoint.

7 Lean Formalization

The current formal route can be summarized as follows:

```
hilbert_right_angle_exists_nondegenerate
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  Exists (fun X : Geo.Point =>
    And (Not (PrimCollinear Geo A C X))
        (HilbertRightAngle Geo A C X))
```

The theorem constructs the line through A, B , an off-line point D , and an angle-copy point E , and then splits on

```
by_cases hCED : Collinear Geo C E D
```

The collinear branch ends with D as the right-angle point. Only the noncollinear branch constructs F and midpoint X and later executes

```
rcases hRayOrder with hCaseD | hCaseE
```

The two ray-order branches establish the same side-configuration biconditional and then share the final angle-addition step.

The next theorem is only a projection. In ASCII-only Lean notation, the logical content can be written equivalently as

```
theorem hilbert_right_angle_exists ... := by
  exact Exists.elim
    (hilbert_right_angle_exists_nondegenerate
      Geo A C B hACB)
    (fun X h => Exists.intro X h.2)
```

Finally, `euclid_proposition_11` calls `hilbert_right_angle_exists` directly and introduces no new geometry.

8 Relationship with Earlier Propositions

8.1 Proposition I.1

Euclid uses I.1 to construct the equilateral triangle DFE . The current Hilbert proof does not call `euclid_proposition_1`; its auxiliary geometry is obtained from Hilbert angle and segment construction and midpoint existence.

8.2 Proposition I.3

Euclid uses I.3 for the layoff $CE \cong CD$ on the opposite ray. The formal reconstruction uses the generic Hilbert segment-construction machinery instead of calling the proposition theorem.

8.3 Proposition I.8

Euclid uses I.8 for SSS on *DCF* and *ECF*. The current formal proof also contains an SSS step, but in the different pair of triangles *CFX* and *CEX*, and it invokes the generic `HilbertSSS` interface rather than `euclid_proposition_8`.

8.4 Transition to Proposition I.12

I.11 constructs a perpendicular at a point already on the given line. I.12 solves the complementary problem from a point outside the line. Together they form the basic perpendicular-construction block used by the subsequent straight-angle propositions.

9 Hilbert and Book Zero Components Used

The active proof of `hilbert_right_angle_exists_nondegenerate` uses, among other things:

- line construction and an off-line point;
- Hilbert III.4 angle construction;
- SameSide/OppositeSide and same-ray transport;
- segment construction on a ray;
- midpoint existence;
- `HilbertSSS`;
- the same-side ray-order theorem;
- line uniqueness and plane-separation lemmas;
- synthetic angle addition.

The angle-construction uniqueness field is returned by the API but is not used in the current active I.11 proof. It should therefore not be shown as a dependency edge of this proposition.

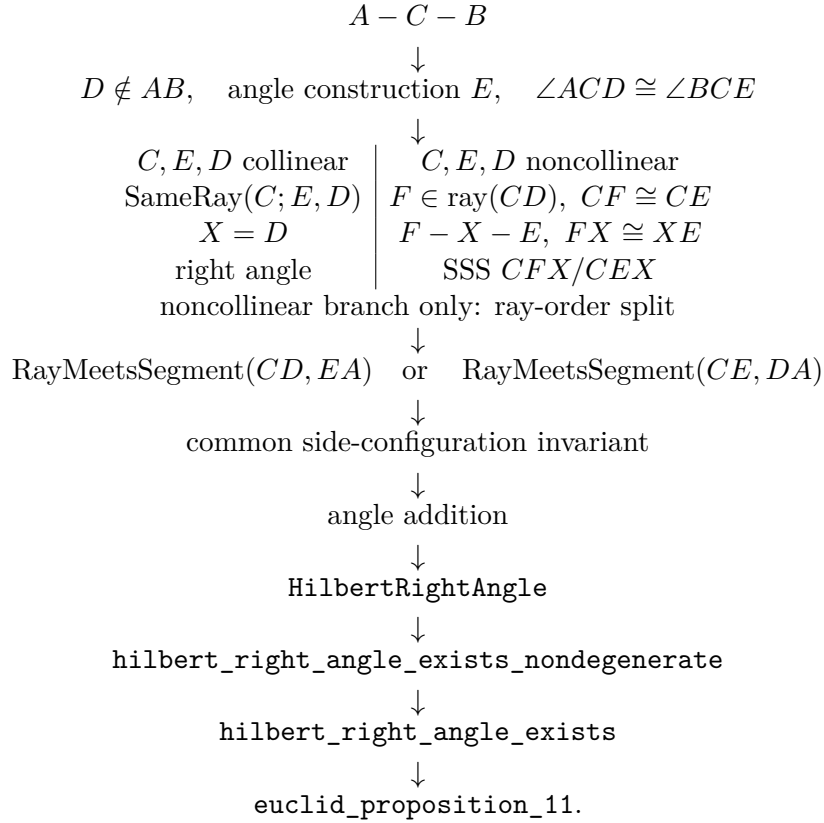
10 Dependency Summary

The classical and formal architectures should be kept distinct.

For Euclid:

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow \text{I.11}.$$

For the current Hilbert reconstruction:



11 Diagrammatic Audit

The preserved Book1R chapter contained one numbered geometric figure: the classical Euclid–Wilkins construction. That figure is retained as Figure 1.

The current production Lean, however, contains two genuine geometric case splits that the preserved diagram set did not record. The first split distinguishes whether the angle-copy point E is collinear with C, D . The second, inside the noncollinear branch, distinguishes which of the two rays CD and CE meets the opposite open segment. These are not tactic-only alternatives: they are different incidence and side-of-line configurations. Figure 2 therefore records them in one four-panel formal map.

No separate figure is added for the final angle-addition step because no new point, line, betweenness, intersection, or side relation appears there. Likewise the uniqueness field returned by angle construction is not diagrammed because it is not consumed by the current proof.

Thus the audited set is

$$1 \text{ classical panel} \quad + \quad 4 \text{ formal panels} \quad = \quad 5 \text{ panels in 2 figures.}$$

12 Conclusion

Proposition I.11 is a clear example of reconstruction rather than literal transcription. Euclid obtains the perpendicular from a symmetric equilateral-triangle construction and I.8. The Hilbert library proves a stronger reusable nondegenerate right-angle existence theorem through angle

construction, segment layoff, midpoint geometry, SSS, plane separation, ray order, and angle addition; two thin wrappers then expose the result as Euclid I.11.

The audited diagrams now reflect both layers without mixing them: one classical construction and one formal four-panel map of the actual current case structure.